



Average Rate and Instantaneous Rate

Today's Learning Objective

I can explain the difference between average rate of change and instantaneous rate of change, and I can use smaller and smaller intervals to estimate a rate at one exact moment.

1

Big Idea

Average rate of change

How fast something changes over an interval. You need two inputs.

Think: average speed for an entire car ride.

Instantaneous rate of change

How fast something changes at one exact moment.

Think: what a speedometer says right now.

Secant line

A line through two points on a curve. Its slope gives an average rate.

Tangent line

A line that matches the curve at one point. Its slope gives an instantaneous rate.

Quick memory check: secant = two points = average rate tangent = one point = instantaneous rate

Average Rate of Change

$$\text{Average rate} = \frac{\text{change in output}}{\text{change in input}} = \frac{f(b) - f(a)}{b - a}$$

For the car: change in position over change in time.

2

Use One Example All the Way Through

Suppose $s(t) = t^2$ gives a car's position in feet after t seconds.

Find the average rate of change from $t = 2$ to $t = 3$.

Step 1. Find the two outputs.	$s(2) = \underline{\hspace{2cm}}$ and $s(3) = \underline{\hspace{2cm}}$
Step 2. Write the average rate as a fraction.	$\text{AROC} = \frac{s(3) - s(2)}{3 - 2} = \frac{9 - 4}{3 - 2} = \underline{\hspace{2cm}}$
Step 3. Simplify and interpret.	The car's position changed at an average rate of $\underline{\hspace{2cm}}$ ft/s.

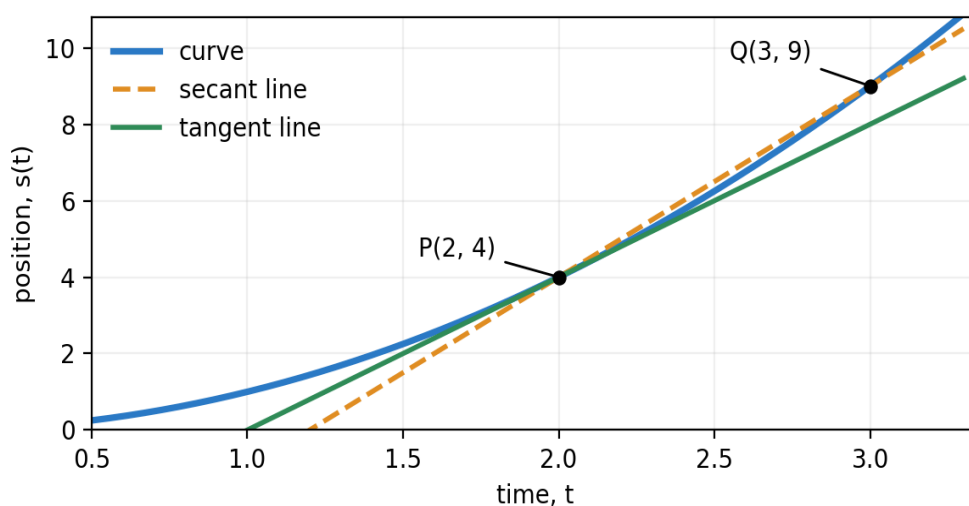


Guided Notes and Practice

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Connect the Graph to the Car Trip

The graph shows the same car trip. Point P is the car at $t = 2$ seconds. Point Q is the car at $t = 3$ seconds. The secant line connects P and Q, so it represents the average speed from 2 to 3 seconds. The tangent line touches the curve at P, so it estimates the speed at exactly 2 seconds.



Secant line

Average speed over the whole chunk from 2 to 3 seconds.

Tangent line

Speed right now at exactly 2 seconds.

Real-world reminder: a secant line is like finding your average speed for part of a trip. A tangent line is like checking the speedometer at one exact instant.



Guided Notes and Practice

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Shrink the Interval

We want the rate at exactly $t = 2$. We cannot use $t = 2$ in both spots because the denominator would be 0.

So we keep one time at 2 and move the other time closer and closer to 2.

3 $\rightarrow$ **2.5** $\rightarrow$ **2.1** $\rightarrow$ **2.01** Smaller interval = better estimate of the rate right now.

Use the same example: $s(t) = t^2$ at $t = 2$.

Time interval	Set up the average rate	Average rate
[2, 3]	$\frac{[3^2 - 2^2]}{[3 - 2]}$	5.00
[2, 2.5]	$\frac{[2.5^2 - 2^2]}{[2.5 - 2]}$	_____
[2, 2.1]	$\frac{[\quad]}{[\quad]}$	_____
[2, 2.01]	$\frac{[\quad]}{[\quad]}$	_____

What number are the average rates getting closer to? _____

So the estimated instantaneous rate at $t = 2$ is _____.

New idea: the number the average rates are getting closer to is the estimated instantaneous rate. In the next class, you can connect this idea to limits.



Guided Notes and Practice

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Try One

Let $f(x) = x^2 + 1$. Estimate the instantaneous rate of change at $x = 1$ by using smaller intervals.

Interval	Average rate
$[1, 1.5]$	_____
$[1, 1.1]$	_____
$[1, 1.01]$	_____

My estimate at $x = 1$ is _____ because the average rates are approaching _____.

6

Exit Check

Two points $\rightarrow$ _____ $\rightarrow$ _____

One moment $\rightarrow$ _____ $\rightarrow$ _____

Why make the interval smaller? _____

From Class

- Average rate uses two inputs and measures change over an interval.
- Instantaneous rate is the rate at one exact moment.
- A secant line matches average rate. A tangent line matches instantaneous rate.
- Making the interval smaller helps average rate get closer to instantaneous rate.